LLIE_F^2: Low Light Image Enhancement by Filters and Frequency domain representations

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1 Abstract

In the frequency domain, the amplitude component of an image primarily governs global luminance and contrast, while the phase component preserves essential structural edges and contours. Existing Fourier-based low-light image enhancement methods often struggle to balance these components, leading to noise amplification in high-frequency regions or structural misalignment. To address these limitations, we propose a dual-modality, two-stage framework termed **LLIE_F**^{2}, which integrates interpretable digital signal processing theories into deep learning.

The framework consists of a Frequency Domain Coarse Reconstruction Stage and a Texture Fine Reconstruction Stage. In the first stage, we explicitly decouple the restoration process: we employ a Luminance Attention Network regulated by a Butterworth Low-Pass Filter to guide amplitude enhancement, preventing noise spikes in dark regions. Simultaneously, we incorporate registered infrared information, stabilized by first-order IIR recursive filters, to correct the phase component and enforce structural priors. In the second stage, we introduce a dual-branch refinement network that combines Fast Fourier Convolutions (FFC) for global consistency with a local texture branch embedded with adaptive FIR (LMS-based) and high-order IIR filters. This hybrid design effectively suppresses noise oscillations while recovering fine-grained details. Experimental results demonstrate that LLIE_F^2 achieves superior visual quality and stability by synergizing classical digital filtering with data-driven representations.

Extensive experiments are performed on the LOL-v2-Real, LOL-v2-Syn, and LSRW-Huawei datasets. Both visual assessments and quantitative benchmarks validate the superiority of our approach, demonstrating that it achieves state-of-the-art performance compared to existing methods.

2 Introduction

Low-light image enhancement (LLIE) is a fundamental component of modern computer vision systems, playing a pivotal role in applications ranging from autonomous driving and surveillance to nocturnal photography. In real-world low-light scenarios, images are degraded by low visibility, severe noise, and color distortion due to the limited photon count received by camera sensors. The primary objective of LLIE is to restore normal exposure and recover latent structural details

while suppressing noise, thereby improving both human visual perception and the performance of downstream machine vision tasks.

The evolution of LLIE techniques can be broadly categorized into non-learning-based and learning-based methodologies. Traditional non-learning approaches, such as Histogram Equalization (HE) and Retinex-based models, rely on statistical properties or physical priors to adjust dynamic range. While computationally efficient, these methods often fail to adapt to complex scene dynamics, resulting in over-enhancement, color artifacts, or amplified noise in extremely dark regions. The advent of deep learning has revolutionized this field, with Convolutional Neural Networks (CNNs) and GANs learning data-driven mappings from low-light to normal-light domains. Recently, researchers have expanded the search space from the spatial domain to the frequency domain. Fourier-based methods leverage the property that the amplitude component primarily encodes global illumination, while the phase component preserves structural information. Additionally, to compensate for the loss of structural cues in the visible spectrum, multi-modal fusion techniques incorporating infrared imagery have emerged as a promising direction, as infrared sensors capture thermal radiation unaffected by illumination conditions. Despite these advancements, significant limitations persist in current state-of-the-art methods. **First**, existing Fourier-based approaches often manipulate amplitude and phase in isolation or simply replicate phase information. This naive treatment neglects the instability of high-frequency components; without explicit constraints, neural networks tend to amplify high-frequency noise alongside details, leading to ringing artifacts and "dirty" textures. **Second**, while infrared images provide robust structural priors, fusing them directly with visible features often introduces modality gaps and spectral jitter, which can degrade the realism of the final output. **Third**, most deep learning models operate as "black boxes," lacking the interpretability and stability of classical signal processing tools. They struggle to distinguish between useful high-frequency texture and random sensor noise, particularly in the fine-grained reconstruction stage.

To address these challenges, we propose a novel dual-modality framework termed **LLIE_F^2**, which synergizes the interpretability of classical digital signal processing (DSP) with the representational power of deep learning. We argue that enhancing low-light images requires not just learning a mapping, but explicitly *filtering* the signal in both frequency and spatial domains. Our method operates in two stages: a **Frequency Domain Coarse Reconstruction Stage** and a **Texture Fine Reconstruction Stage**. In the first stage, we introduce explicit digital filters—specifically *Butterworth Low-Pass Filters* and *first-order IIR recursive filters*—to rigorously stabilize the luminance attention and infrared phase priors. This ensures that the global illumination is corrected without amplifying noise. In the second stage, we design a local texture branch equipped with *Adaptive FIR filters (optimized via LMS)* and *high-order IIR filters*. These modules act as differentiable signal processors that adaptively smooth out noise while preserving genuine micro-textures.

The main contributions of this work are summarized as follows:

- We propose LLIE_F^2, an end-to-end framework that integrates registered infrared information to guide the restoration of visible low-light images, effectively solving the problem of structural information loss in extreme darkness.
- We seamlessly incorporate interpretable digital filters (Butterworth, IIR, and Adaptive FIR) into deep neural networks. To the best of our knowledge, this is one of the first attempts to use explicit DSP theories to stabilize frequency-domain features and spatial texture responses in LLIE tasks.

- We design a dual-stage architecture that decouples global illumination correction (via Amplitude-Phase modulation) from local detail refinement, ensuring global consistency while maximizing local texture fidelity.
- Extensive experiments on the LOL-v2-Real, LOL-v2-Syn, and LSRW-Huawei datasets demonstrate that our approach achieves state-of-the-art performance in both quantitative metrics and visual quality.

3 Literature Review

Deep learning has become a cornerstone of modern computer vision, demonstrating remarkable efficiency in tasks ranging from object detection to image restoration. In the specific domain of Low-Light Image Enhancement (LLIE), deep learning techniques have emerged as the dominant research trend, significantly outperforming traditional signal processing methods in improving visibility and recovering details from dark inputs. Contemporary research primarily bifurcates into two methodologies: Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs).

3.1 CNN-Based Enhancement Frameworks

Convolutional Neural Networks (CNNs) generally operate as supervised learning models in the context of image enhancement. They function by learning a complex, non-linear mapping between low-light inputs and high-quality ground truth images. The architecture typically comprises convolutional layers for local feature extraction, pooling layers for parameter reduction, and fully connected layers for reconstruction. CNNs are favored for their end-to-end training capabilities and robust feature learning; however, they may suffer from overfitting if training data is scarce and can sometimes neglect global image consistency.

3.1.1 Physical Model-Based Approaches

A significant branch of CNN research incorporates physical constraints to guide the enhancement process. The foundational work by Lore et al. [1] introduced LLNet, utilizing a deep autoencoder to simultaneously denoise and brighten images. This paved the way for more complex architectures, such as the Multi-Branch Low-Light Enhancement Network (MBLLEN) proposed by Lv et al. [2], which extracts features across different distinct modules—enhancement, fusion, and extraction—to preserve signal integrity.

Subsequent studies focused on specific degradation issues. To combat the gradient vanishing problem common in deep networks, Tao et al. [3] developed LLCNN, employing multi-scale feature maps and Structural Similarity Index (SSIM) loss to maintain textural fidelity. Similarly, decomposition strategies were adopted by Xu et al. [4], who processed high-frequency details and low-frequency content separately to suppress noise while enhancing contrast. Furthermore, hybrid architectures combining Encoder-Decoder structures with Recurrent Neural Networks (RNNs) have been proposed to simultaneously address edge preservation and content enhancement [5]. More recently, multi-stage fusion methods, such as the Edge-Enhanced Multi-Exposure Fusion Network (EEMEFN), generate synthetic exposure sequences to fuse optimal pixel information [6].

3.1.2 Non-Physical Model-Based Approaches

Alternatively, researchers have developed data-driven approaches that do not strictly adhere to physical image formation models, or they integrate classical theories like Retinex implicitly. Gharbi et al. [7] revolutionized real-time processing by reformulating the bilateral filter within a deep learning framework.

Retinex theory remains a strong influence even in non-physical models. The Multi-Scale Retinex Network (MSR-net) [8] and Deep Retinex Decomposition [9] leverage CNNs to estimate illumination and reflectance components essentially learning to map dark images to normal brightness via physics-inspired decomposition. Li et al. [10] proposed LightenNet, which predicts a light map to guide the enhancement, although it occasionally struggles with complex real-world scenes. To improve contrast learning, Cai et al. [11] utilized multi-exposure data to train deep enhancers. Other innovations include integrating guided filters for high-frequency preservation [12] and employing RNNs with non-local mean filtering to utilize spatial information for denoising [13].

3.2 GAN-Based Enhancement Frameworks

Generative Adversarial Networks (GANs) represent a paradigm shift towards unsupervised or semi-supervised learning. A GAN consists of a generator that synthesizes candidate images and a discriminator that evaluates their authenticity. In LLIE, the generator aims to produce enhanced images that are statistically indistinguishable from high-quality reference images. While GANs excel at generating realistic textures and avoiding the blurry results often associated with Mean Squared Error (MSE) loss in CNNs, they are notoriously difficult to train, suffering from instability and mode collapse.

[Image of Generative Adversarial Network architecture]

3.2.1 Condition-Based and Weakly Supervised Methods

To mitigate the reliance on perfectly paired training data, which is difficult to capture in real-world scenarios, Jiang et al. [14] proposed EnlightenGAN. This unsupervised framework employs a U-Net generator with a dual global-local discriminator and self-regularization losses to maintain content consistency without paired supervision.

Similarly, Ignatov et al. moved from strong supervision [15] to weakly supervised learning with WESPE [16]. This method uses a transitivity-based GAN to map mobile camera images to DSLR quality, enforcing content consistency via VGG19 feature losses rather than pixel-wise correspondence. Further applications of conditional GANs extend to nighttime enhancement [17] and general photo restoration [18], proving that adversarial training can effectively hallucinate missing details in low-light conditions.

3.2.2 Cycle Consistency-Based Methods

The introduction of CycleGAN facilitated unpaired image-to-image translation by enforcing cycle consistency—ensuring an image translated to a target domain can be translated back to the source domain. Zhu et al. [19] pioneered this for general style transfer, which was adapted for LLIE by Fu et al. [20] in LE-GAN. LE-GAN incorporates illumination-aware attention modules and identity-invariant loss to handle noise and color bias specific to low-light scenarios.

Optimization within the GAN framework has also been explored; Yan et al. [21] utilized an optimized enhancement module to generate latent space representations. In the medical imaging domain, You et al. [22] proposed Cycle-CBAM, integrating the Convolutional Block Attention Module into a CycleGAN structure to enhance retinal images, demonstrating the versatility of circular consistency constraints in preserving anatomical details during enhancement.

3.3 Multi-Modal and Infrared-Visible Image Fusion

While single-spectrum low-light enhancement focuses on recovering information from visible light signals, performance often degrades in extremely dark environments where photon counts are negligible. To address this, recent research has pivoted towards multi-modal fusion, specifically integrating Infrared (IR) and Visible (VI) imagery. Infrared sensors capture thermal radiation, providing crucial structural and semantic information that remains unaffected by illumination conditions [23, 24].

3.3.1 Translation and Data Scarcity

A primary bottleneck in deep learning-based fusion is the scarcity of aligned visible-infrared image pairs. To circumvent this, researchers have employed image-to-image translation techniques. Kniaz et al. [25] and Nyberg et al. [26] utilized Generative Adversarial Networks (GANs) to synthesize thermal data from visible spectrum inputs, enabling the training of robust fusion models even in the absence of paired ground truth. Further refining this approach, Lee et al. [27] introduced edge-guided translation models designed to preserve high-frequency spatial textures during the domain transfer process, ensuring that the synthesized IR images retain the structural fidelity required for effective fusion [28].

3.3.2 Feature Decomposition and Fusion Strategies

Moving beyond simple concatenation, recent architectures employ sophisticated feature decomposition. Zhao et al. [29] proposed Cddfuse, a correlation-driven dual-branch network that decomposes features into modality-specific and modality-shared components. This allows the model to selectively fuse distinct information—thermal targets from IR and texture details from VI—without introducing noise. Similarly, meta-learning approaches like MetaFusion [24] embed feature embedding from object detection tasks to guide the fusion process, ensuring that the enhanced images are not only visually pleasing but also semantically informative for downstream vision tasks.

3.4 Frequency Domain and Fourier-Based Methods

Traditional CNN-based methods predominantly operate in the spatial domain. However, spatial operations often have a limited receptive field and struggle to differentiate between low-frequency illumination changes and high-frequency noise. This has led to a surge of interest in Fourier-based LLIE methods, which leverage the frequency domain to process global image characteristics more efficiently.

3.4.1 Amplitude and Phase Disentanglement

The core premise of frequency-based enhancement relies on the property that the amplitude spectrum typically encodes low-level statistics (such as brightness and contrast), while the phase spec-

trum preserves high-level structural information. Huang et al. [30] exploited this by designing a spatial-frequency interaction network that specifically manipulates the amplitude component for exposure correction while preserving phase information to maintain structural integrity. This disentanglement allows for targeted illumination enhancement without the artifacts common in spatial-only methods.

3.4.2 Global Context and Efficiency

Fourier transforms offer a mechanism to capture global context with minimal computational overhead compared to non-local spatial attention blocks. Wang et al. [31] introduced FourLLIE, demonstrating that global feature extraction can be achieved in the Fourier domain without significantly increasing model parameters. By learning a Signal-to-Noise Ratio (SNR) aware mapping in the frequency domain, the model can effectively distinguish between signal and noise, a critical capability for low-light scenes. Furthermore, for ultra-high-definition (UHD) content, Li et al. [32] proposed UHDFour, which utilizes the observation that amplitude components exhibit similarity across different resolutions, allowing for efficient enhancement of 4K imagery by processing downsampled frequency components.

4 Methodology

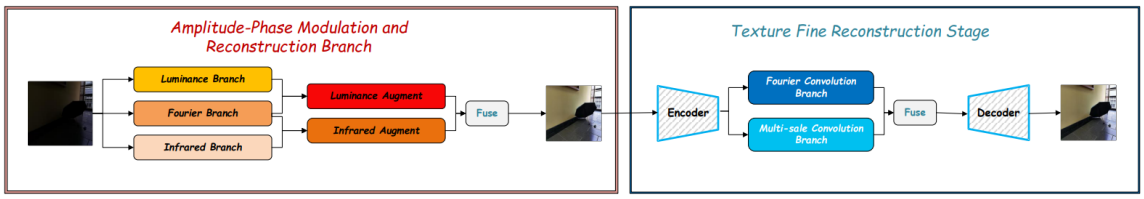


Figure 1: The overall structure of **LLIE_F^2** , comprising frequency-domain coarse reconstruction and spatial-domain texture refinement stages, integrated with diverse filters for improved spectral stability and local detail enhancement.

We propose a dual-modality, two-stage framework for low-light image enhancement, termed LLIE_F^2, as shown in Figure 1. Given a low-light visible image I_{low} and its registered infrared image I_{ir}, the model learns an end-to-end mapping to produce an enhanced output $\hat{I}$. The overall network consists of two parts:

(1) **Frequency Domain Coarse Reconstruction Stage** for global reconstruction, which incorporates a luminance attention map and infrared information to extract and enhance features in the frequency domain by separately modeling the amplitude and phase components. During this process, we introduce multiple interpretable filters (e.g., FIR/IIR filters, a Butterworth low-pass filter, and a high-pass/low-pass/residual combination) to stabilize the frequency-domain representation, and output the first-stage result I_{s1}, providing a more reasonable global exposure and structural basis for subsequent refinement.

(2) **Texture Fine Reconstruction Stage** for local detail recovery, which adopts dual-branch feature modeling: one branch leverages modules such as FFC to strengthen long-range dependencies and global consistency, while the other branch employs multi-scale convolutions and filtering

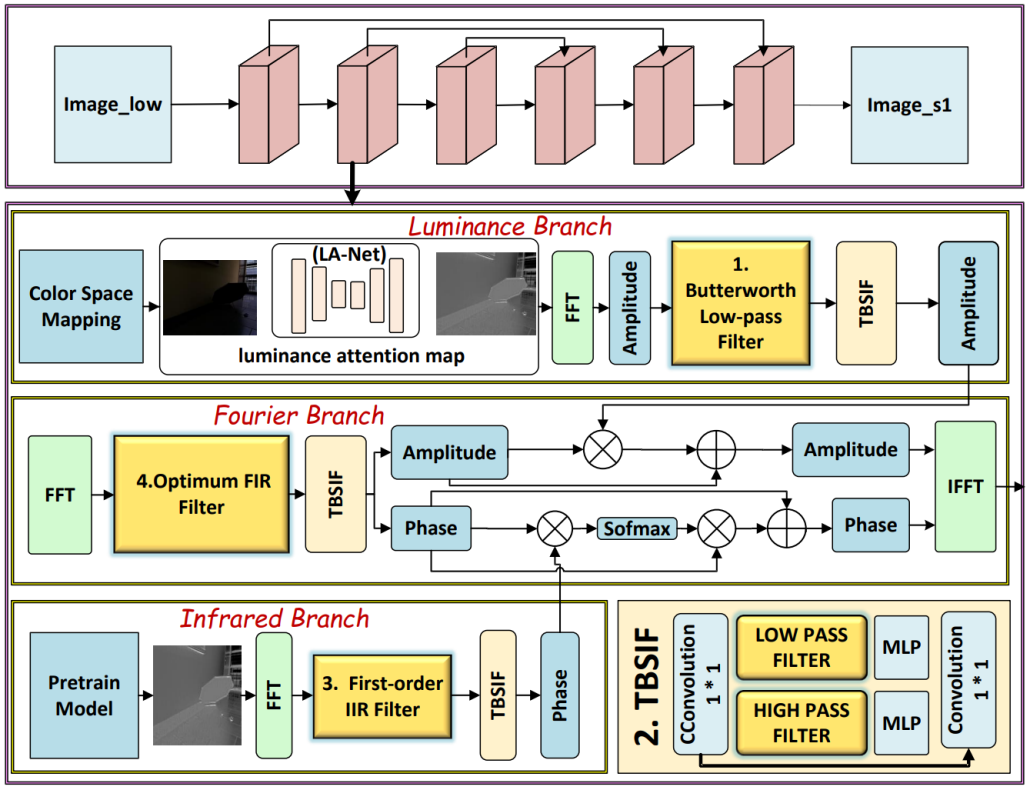


Figure 2: The overall structure of Frequency Domain Coarse Reconstruction Stage in **LLIE_F**^{2}.

to enhance local texture representation. The two branches are adaptively integrated via channel-attention fusion, and the final enhanced image $\hat{I}$ is reconstructed through progressive upsampling.

4.1 Frequency Domain Coarse Reconstruction Stage

As shown in Figure.2. Given a low-light visible input image $I_{low} \in \mathbb{R}^{3\times H\times W}$ and a registered infrared image $I_{ir} \in \mathbb{R}^{3\times H\times W}$, this stage performs frequency-domain modeling of both amplitude and phase, and generates the first-stage output I_{s1}. The key motivation is that the amplitude component primarily governs the global brightness/contrast distribution, while the phase component carries most structural and edge information. Therefore, instead of enhancing amplitude alone or naively replicating phase, we incorporate a set of *interpretable digital filters* together with attention-based modulation to stabilize, denoise, and selectively strengthen informative components on dedicated branches.

This stage is implemented as a stack of several (six in our implementation) identical frequency-domain processing modules connected in series, with feature communication realized by cross-layer concatenation. Each module takes the visible feature, the infrared feature, and the luminance attention map from the previous module, and outputs updated versions of these three. Finally, the enhanced amplitude and phase are transformed back to the spatial domain via inverse Fourier

transform, and a residual connection produces I_{s1}.

4.1.1 Luminance Branch and Butterworth Low-Pass Filtering

Low-light enhancement is prone to under-enhancement in dark regions and over-exposure in bright regions, and noise is typically more pronounced in high-frequency components. We first extract the luminance channel Y_{low} in the $YCrCb$ space and predict a luminance attention map using a U-Net-style luminance attention network (LA-Net):

$$L_{att} = \sigma\big(\text{LA-Net}(Y_{low})\big), \qquad L_{att} \in [0,1]^{1\times H\times W}, \tag{1}$$

where $\sigma(\cdot)$ denotes the Sigmoid function. During training, a luminance supervision target can be constructed based on the relative discrepancy between the input luminance and the ground-truth luminance Y_{gt}:

$$L_{att}^{gt} = \text{clip}\left(\frac{|Y_{low} - Y_{gt}|}{|Y_{gt}| + \epsilon},\, 0,\, 1\right). \tag{2}$$

After obtaining $\tilde{L}_{att}$, we transform the spatial attention map into the frequency domain to extract global structural features. This is achieved by applying the Fast Fourier Transform (FFT), which projects the representation into the Fourier domain:

$$X = \mathcal{F}(\tilde{L}_{att}), \tag{3}$$

where $\mathcal{F}(\cdot)$ denotes the 2D FFT operation. This amplitude representation A_{lum} serves as the initial luminance guidance.

To ensure the attention map is smooth and stable while suppressing noise-induced local spikes, we apply a *Butterworth Low-pass Filter* (BLPF) to L_{att} in the frequency domain. Its frequency response is defined as:

$$H_{bw}(D) = \frac{1}{1 + \left(\frac{D}{D_0}\right)^{2n}}, \tag{4}$$

where D is the normalized radial distance to the origin on the frequency plane, D_0 is the cutoff frequency (smaller D_0 yields stronger smoothing), and n is the filter order (larger n yields a sharper transition band).

To further refine the filtered features and balance high-frequency details with smoothed components, the signal is processed through a specialized Tri-Branch Spectral Integration Filter (TBSIF) block. This block is embedded within a processing sequence consisting of a 1×1 convolution, the TBSI module, and a LeakyReLU activation.

The TBSI block decomposes the input feature map x into three parallel paths:

- **High-pass Filter Branch:** Captures fine textures and edge gradients using a fixed Laplacian kernel ∇^2. The weights are frozen.
- **Low-pass Filter Branch:** Employs a learnable Finite Impulse Response (FIR) filter to maintain structural consistency and low-frequency stability.
- **Residual Filter Branch:** A standard identity skip connection that preserves the original feature characteristics.

The outputs of these three branches are integrated via summation:

$$x_{out} = \text{Branch}_{high} + \text{Branch}_{low} + \text{Branch}_{res}. \tag{5}$$

Finally, the magnitude of the refined frequency domain feature $A_{lum} = |x_{out}|$ is calculated as the final output of the luminance branch. This representation is then forwarded to the subsequent Fourier reconstruction stage as the optimized luminance guidance, and the luminance branch terminates.

4.1.2 Infrared Branch and IIR Recursive Filtering

Infrared images typically provide more reliable contours and structural cues in dark conditions. We transform the infrared input to the frequency domain and extract its phase information as a structural prior:

$$\mathbf{X}_{ir}(u,v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} I_{ir}(h,w)\cdot e^{-j2\pi\left(\frac{uh}{H}+\frac{vw}{W}\right)}, \tag{6}$$

where $\mathbf{X}_{ir}(u,v)$ represents the spectral component in the frequency domain at coordinates (u,v), and $I_{ir}(h,w)$ denotes the pixel intensity of the spatial input image at position (h,w). H and W refer to the height and width of the image, respectively, while $e^{-j2\pi(\cdot)}$ serves as the complex exponential basis function of the Fourier transform.

To further suppress random fluctuations and stabilize cross-modal fusion, we apply a *first-order IIR recursive filter* (implemented as a width-wise recursion) to the infrared feature $\mathbf{X}_{ir}(u,v)$:

$$y[n] = \alpha x[n] + (1-\alpha)y[n-1], \qquad \alpha \in (0,1), \tag{7}$$

where α controls smoothing strength (smaller α yields stronger smoothing).

The IIR filter is computationally efficient and accumulates history, effectively reducing jitter and preserving the overall trend, which is particularly important for stable phase-attention computation.

After that, the infrared image signal is processed by a Tri-Branch Spectral Integration Filter (TBSIF) block we defined before.

Finally, we extract the phase component from the representation processed by the first-order IIR recursive filter and the TBSIF. This is denoted as:

$$P_{inf} = \angle \mathbf{X}_{ir}(u,v). \tag{8}$$

This phase information P_{inf} is subsequently employed to enhance phase characteristics in the frequency domain, thereby improving structural retention and edge definition.

4.1.3 Amplitude-Phase Modulation and Reconstruction

With the optimized luminance guidance A_{lum} (filtered by BLPF and TBSIF) and the structural phase prior P_{inf} (stabilized by IIR and TBSIF) obtained from the auxiliary branches, the main branch performs the core restoration of the low-light input I_{low}. First, the input visible image I_{low} is projected into the Fourier domain via FFT to obtain its complex spectral representation $\mathbf{X}_{low}$:

$$\mathbf{X}low = \mathcal{F}(Ilow) \in \mathbb{C}^{H\times W}. \tag{9}$$

Directly decoupling the raw spectral components often results in amplitude spikes and phase jitter due to high-frequency noise. To mitigate this, we propose a pre-decomposition spectral smoothing strategy.

Spectral Stabilization via Optimum FIR Filter Before extracting the amplitude and phase, we apply a *finite impulse response (FIR)* digital low-pass filter directly to the complex spectrum $\mathbf{X}_{low}$. This process suppresses outlier fluctuations in the frequency domain while preserving the dominant spectral structure. The filtering is performed on both the real and imaginary components:

$$\tilde{\mathbf{X}}low = \mathrm{FIR}(\mathbf{X}low). \tag{10}$$

In practice, we employ the *optimum FIR design method*, which yields a fixed, symmetric 3×3 smoothing kernel (least-squares-optimal in the mean-square sense):

$$K_{\mathrm{FIR}} = \frac{1}{16}\begin{bmatrix}1 & 2 & 1 & 4 & 2 & 2 & 1\end{bmatrix}. \tag{11}$$

This kernel is implemented as a parameter-free depthwise convolution applied to the spectral feature map. Finally, the stabilized amplitude $\tilde{A}_{low}$ and phase $\tilde{P}_{low}$ are decoupled from the filtered spectrum $\tilde{\mathbf{X}}_{low}$:

$$\tilde{A}_{low} = |\tilde{\mathbf{X}}low|, \qquad \tilde{P}_{low} = \angle\tilde{\mathbf{X}}low. \tag{12}$$

By filtering the complex spectrum prior to decomposition, we ensure that the subsequent amplitude and phase representations are numerically stable and free from decomposition artifacts.

Amplitude Modulation via Filtered Luminance The smoothed amplitude component $\tilde{A}_{low}$ typically suffers from low visibility. We modulate $\tilde{A}_{low}$ using the learned luminance guidance A_{lum}. Crucially, the BLPF applied in the luminance branch restricts the guidance to the low-to-mid frequency range, preventing the enhancement process from amplifying noise in dark regions. The modulated amplitude is formulated as:

$$A_{rec} = \mathcal{M}_{amp}(\tilde{A}_{low}, A_{lum}), \tag{13}$$

where $\mathcal{M}_{amp}$ denotes the amplitude modulation function.

Phase Correction via Stabilized Infrared Structure The smoothed phase component $\tilde{P}_{low}$ may still lack clear structural definition. We utilize the infrared phase prior P_{inf} to correct $\tilde{P}_{low}$. The IIR filter and TBSIF in the infrared branch ensure that the injected phase information is spatially smooth and robust. The phase refinement is defined as:

$$P_{rec} = \mathcal{M}_{phase}(\tilde{P}_{low}, P_{inf}), \tag{14}$$

where $\mathcal{M}_{phase}$ represents the phase fusion operation.

Inverse Reconstruction Finally, the reconstructed amplitude A_{rec} and phase P_{rec} are recombined using Euler's formula to restore the complex frequency representation, which is then mapped back to the spatial domain via the Inverse Fast Fourier Transform (IFFT):

$$I_{s1} = \mathcal{F}^{-1}\left(A_{rec} \cdot e^{j \cdot P_{rec}}\right). \tag{15}$$

This coarse result I_{s1} possesses corrected global illumination and restored structural integrity, serving as a robust foundation for the subsequent Texture Fine Reconstruction Stage.

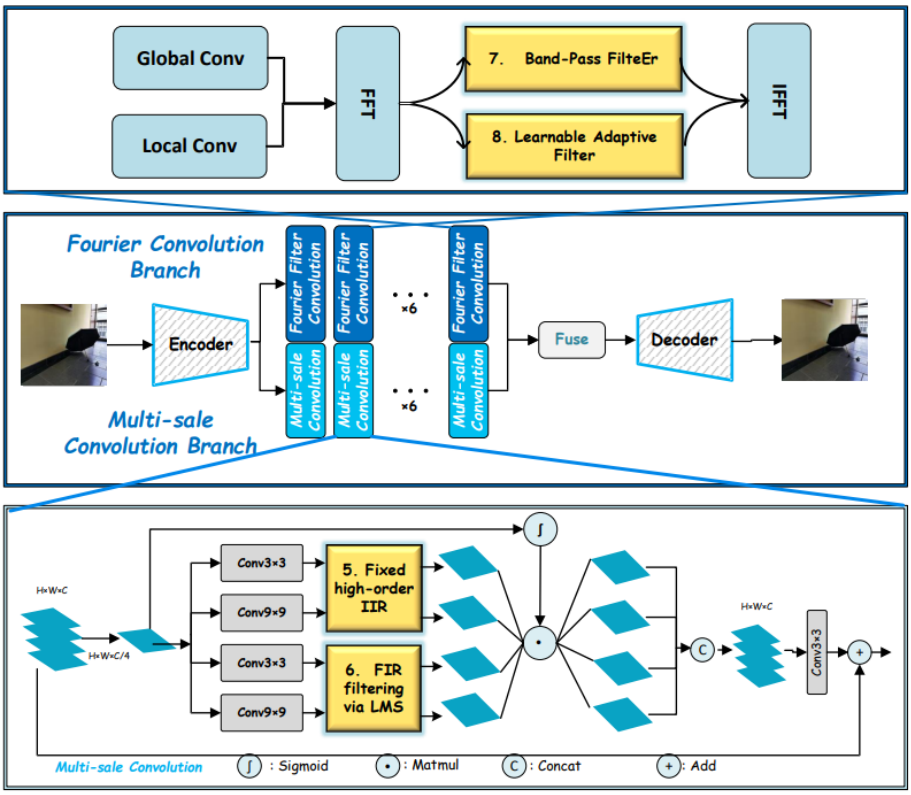


Figure 3: The overall structure of Texture Fine Reconstruction Stage in **LLIE_F**^{2}.

4.1.4 Summary: The Role of Filters in the First Coarse Stage

- **Butterworth low-pass (on the luminance branch):** removes high-frequency noise from the attention guidance signal in the frequency domain, making "where to enhance" smoother and more reliable.
- **Tri-Branch Spectral Integration Filter (TBSIF):** preserves edges while suppressing noise during frequency-branch mappings, avoiding blur from pure low-pass filtering and noise amplification from pure high-pass filtering.
- **Optimum FIR (on amplitude and phase branches):** uses a deterministic, parameter-free smoothing kernel to reduce amplitude spikes and phase jitter that can cause reconstruction artifacts, improving training stability.
- **First-order IIR (on the infrared branch):** suppresses infrared feature jitter via recursive smoothing, stabilizing cross-modal attention computation and strengthening phase-based structural guidance.

4.2 Texture Fine Reconstruction Stage

While the Fourier reconstruction stage provides a globally corrected exposure and a structurally consistent intermediate result I_{s1}, fine-grained textures and local details may still be insufficient. Therefore, we introduce a second stage, termed the *Texture Fine Reconstruction Stage*, which refines local textures and restores spatial details in the spatial domain while preserving global consistency. As shown in Figure 3 This stage takes the concatenation of the low-light input and the first-stage output as input, and predicts a residual refinement to produce the final enhanced image $\hat{I}$.

4.2.1 Stage Input and Multi-Scale Encoder

Let $I_{low} \in \mathbb{R}^{3\times H\times W}$ be the low-light visible image and $I_{s1} \in \mathbb{R}^{3\times H\times W}$ be the output of the Fourier reconstruction stage. We concatenate them along the channel dimension:

$$X_0 = \mathrm{Cat}(I_{low}, I_{s1}) \in \mathbb{R}^{6\times H\times W}. \tag{16}$$

To build a compact latent representation and enlarge the receptive field for texture reasoning, we adopt a strided encoder with progressive downsampling:

$$X_1 = \phi(\mathrm{Conv}_{3\times 3}(X_0)), \quad X_2 = \phi(\mathrm{Conv}_{3\times 3,s=2}(X_1)), \quad X_3 = \phi(\mathrm{Conv}_{3\times 3,s=2}(X_2)), \tag{17}$$

where $\phi(\cdot)$ denotes LeakyReLU activation. The feature X_3 serves as the shared input to a dual-branch refinement trunk.

4.2.2 Dual-Branch Refinement: Global Spectral Branch and Local Texture Filtering Branch

To simultaneously model global consistency and fine local textures, we employ two parallel branches operating on X_3.

(1) Global spectral-consistency branch. The first branch is composed of stacked Enhanced FFC residual blocks. While each block performs the standard decomposition of features into local and global channel groups to enable long-range dependency modeling via spectral transformers, we further augment it with a Dual-Path Spectral Refinement mechanism to ensure frequency stability. Specifically, parallel to the FFC transformation, we introduce two complementary filters:

(a) A **Frequency-Domain Band-Pass Filter (BPF)**, which performs FFT to explicitly isolate mid-frequency structural components. It suppresses both low-frequency haze remnants and high-frequency noise using a Butterworth-style transfer function H_{bp}, computed as $Y_{bp} = \mathcal{F}^{-1}(\mathcal{F}(X) \odot H_{bp})$.

(b) A **Learnable Adaptive Filter**, implemented via a depthwise 3×3 convolution. Unlike fixed filters, this module optimizes its spatial weights during training to capture data-dependent spectral patterns and refine local feature consistency.

The outputs of these two filters are injected into the main FFC stream as residual components. Denote the output of this branch after N blocks as:

$$F_{\mathrm{ffc}} = \mathcal{T}_{\mathrm{FFC}}(X_3). \tag{18}$$

This design not only compensates for large-scale illumination structures but also actively suppresses frequency artifacts through this explicit dual-filtering strategy.

(2) Local texture filtering branch (multi-scale convolution + digital filters). The second branch explicitly emphasizes local texture restoration via a *MultiConvBlock* stack, which combines multi-scale depthwise convolutions with *interpretable digital filters*. Denote the output after N blocks as:

$$F_{\text{tex}} = \mathcal{T}_{\text{MCB}}(X_3). \tag{19}$$

Each MultiConvBlock begins with channel reduction to produce a compact texture carrier:

$$Z = \phi(\text{Conv}_{1\times1}(X)), \qquad Z \in \mathbb{R}^{\frac{C}{4}\times h\times w}. \tag{20}$$

Then, multi-scale depthwise convolutions with different kernel sizes (e.g., 3×3 and 9×9) extract texture responses at different spatial extents:

$$U_i = \phi(\text{DWConv}_{k_i\times k_i}(Z)), \qquad i = 1,\ldots,H, \tag{21}$$

where DWConv denotes depthwise convolution, and $k_i \in \{3, 9\}$ in our design.

Digital filtering (FIR/IIR) for texture stabilization. To stabilize the extracted texture responses and suppress noise-like oscillations, we apply branch-specific digital filters to U_i:

$$\bar{U}_i = \mathcal{D}_i(U_i). \tag{22}$$

Concretely, we use two types of interpretable filters:

(a) Adaptive FIR filtering via LMS (for fine texture branches). For selected branches, we implement an adaptive finite impulse response (FIR) filter optimized online via the LMS rule. For a 1D tap vector of length K along the horizontal direction (applied per channel and per row), the filtering and update are:

$$\hat{y}[n] = \mathbf{w}^\top[n]\mathbf{x}[n], \tag{23}$$

$$e[n] = d[n] - \hat{y}[n], \tag{24}$$

$$\mathbf{w}[n{+}1] = \mathbf{w}[n] + \mu\, e[n]\mathbf{x}[n], \tag{25}$$

where $\mathbf{x}[n]$ is the local K-tap vector, $\mathbf{w}[n]$ are the adaptive taps, μ is the step size, and the desired signal $d[n]$ is set as the current sample (identity target) to realize self-adaptive smoothing. This LMS-based adaptive FIR enables data-dependent suppression of high-frequency noise while preserving meaningful micro-textures.

(b) Fixed high-order IIR smoothing (for robust texture branches). For the remaining branches, we adopt a fixed high-order IIR filter along the horizontal direction:

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + b_1y[n-1] + b_2y[n-2], \tag{26}$$

where $\{a_0, a_1, a_2, b_1, b_2\}$ are fixed coefficients. Compared with FIR, the IIR filter provides stronger recursive smoothing with low computational cost, which is beneficial for stabilizing responses and preventing ringing artifacts in challenging low-light conditions.

Gated modulation and multi-branch fusion. After digital filtering, we apply an element-wise modulation to adaptively gate the branch response using the reduced feature Z:

$$V_i = \bar{U}_i \odot \sigma(Z), \tag{27}$$

where $\sigma(\cdot)$ is Sigmoid. We concatenate all modulated branches and fuse them using a 1×1 convolution, followed by a squeeze-and-excitation (SE) unit for channel re-weighting:

$$V = \mathrm{Cat}(V_1, \ldots, V_H), \quad V_f = \mathrm{SE}(\mathrm{Conv}_{1\times1}(V)). \tag{28}$$

Finally, a residual connection produces the MultiConvBlock output:

$$\hat{I} = X + V_f. \tag{29}$$

Through this design, the local texture branch explicitly embeds *adaptive FIR* and *high-order IIR* filtering into the feature extraction pipeline, yielding stable and detail-preserving texture representations.

4.2.3 Discussion: Why Explicit Digital Filters Work

This stage explicitly integrates interpretable filters into deep feature processing:

- **Adaptive FIR (LMS):** provides data-dependent smoothing and noise suppression, helping recover fine textures without amplifying low-light noise.
- **High-order IIR:** offers efficient recursive stabilization to reduce oscillations and ringing, improving robustness in extremely dark regions.
- **Band-pass Filter + learnable adaptive Filter:** Built upon the global modeling of Fast Fourier Convolution (FFC), this branch achieves a synergistic enhancement of global consistency and spectral stability through parallel fixed band-pass filtering and learnable adaptive filtering.

Together, these designs enable the second stage to perform stable and detail-preserving texture refinement on top of the globally corrected first-stage output.

5 Experiment

5.1 Experiment Settings

To verify the effectiveness of LLIE_F^2, this study conducted evaluations on three mainstream low-light image enhancement (LLIE) benchmark datasets: LOL-v2-Real, LOL-v2-Syn, and LSRW-Huawei. Specifically, LOL-v2-Real consists of 789 image pairs (689 for training and 100 for testing) captured in real-world scenes by adjusting exposure times and ISO settings. Compared to the original LOL dataset, this version offers a significant increase in both scale and scene diversity. LOL-v2-Syn contains 1,000 image pairs (900 for training and 100 for testing) synthesized by modeling the luminance distribution of low-light and normal-light images. The LSRW-Huawei dataset, collected using various imaging devices, provides 3,150 training pairs and 20 test pairs.

The network weights are updated using the ADAM optimizer with an initial learning rate of 4.0×10^{-4}. To ensure a stable training trajectory and precise convergence, a multi-step learning rate scheduling strategy is adopted, which systematically reduces the learning rate at predefined intervals. The model is trained with a mini-batch size of 8, and the entire optimization procedure spans a total of $100,000$ iterations until the loss function reaches a plateau.

We use three metrics to evaluate the performance of our model: PSNR, SSIM, and LPIPS.

Specifically: PSNR (Peak Signal-to-Noise Ratio): PSNR is the most commonly used metric to measure image distortion, primarily based on the Mean Squared Error (MSE) between pixels. In low-light enhancement tasks, a higher PSNR value (measured in dB) indicates that the enhanced image is closer to the reference image in terms of pixel values, signifying better noise suppression. Its calculation formula is typically expressed as:

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \tag{30}$$

SSIM (Structural Similarity Index): Unlike PSNR, which only focuses on pixel differences, SSIM simulates the sensitivity of the human visual system to illumination, contrast, and structural information. Its value ranges from 0 to 1, where a value closer to 1 indicates that structural information (such as object contours and texture details) is preserved more completely with less distortion.

LPIPS (Learned Perceptual Image Patch Similarity): This is a newer metric based on deep learning, often referred to as perceptual loss. It extracts features through a pre-trained deep convolutional network to calculate the distance between the enhanced image and the ground truth image in the feature space. A lower LPIPS value indicates that the image is perceptually closer to natural and realistic effects in terms of human visual sensation, reflecting human aesthetic preferences more effectively than the former two metrics.

5.2 Comparison with Current Methods

To demonstrate the effectiveness of our approach, particularly the design of the various filters, we conduct a comprehensive comparison between our method and existing state-of-the-art (SOTA) algorithms. Our benchmarks include NPE [33], LIME [34], SRIE [35], Kind [36], MIRNet [37], Kind++ [38], SGM [39], FECNet [30], HDMNet [40], SNR-Aware [41], Bread [42], and FourLLIE [31].

The quantitative results on three benchmarks are summarized in Table 1. As observed, our proposed LLIE_F^2 consistently achieves the best performance across all datasets and metrics, denoted in red.

Performance on LOL-v2 Datasets. On the LOL-v2-Real dataset, which represents complex real-world degradation, our method achieves a PSNR of 22.64 dB and an SSIM of 0.8589, outperforming the recent FourLLIE and SNR-Aware. Specifically, compared to these leading methods, our model obtains absolute PSNR gains of 0.30 dB and 1.16 dB, respectively. Furthermore, in terms of SSIM, we exceed the SNR-Aware method by a margin of 0.0111. Notably, we achieve the lowest LPIPS score of 0.0520, indicating that our enhanced images are perceptually closer to the ground truth. This score represents a reduction of 0.0053 points compared to the second-best FourLLIE, highlighting the superior perceptual quality.

On the LOL-v2-Syn dataset, the advantage of our method is even more pronounced. We achieve a PSNR of 25.83 dB, surpassing the second-best method (FourLLIE) by a significant margin of 1.18 dB. Additionally, regarding structural fidelity, we outperform FourLLIE and SNR-Aware by absolute values of 0.0122 and 0.0045 in SSIM, respectively. This substantial improvement suggests that our end-to-end training strategy and frequency-based filter design can effectively learn the mapping between low-light and normal-light distributions.

Table 1: **We will conduct a detailed comparison of our method with the previous State of The Art methods on three datasets. Top: Results on LOL-v2-Real and LOL-v2-Syn. Bottom: Results on LSRW-Huawei. Best in red.**

Methods	LOL-v2-Real			LOL-v2-Syn		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
NPE	17.33	0.4642	0.2359	16.60	0.7781	0.1079
LIME	15.24	0.4190	0.2203	16.88	0.7578	0.1041
SRIE	14.15	0.5524	0.2160	14.50	0.6640	0.1484
Kind	20.01	0.8412	0.0813	22.62	0.9041	0.0515
MIRNet	22.11	0.7942	0.1448	22.52	0.8997	0.0568
Kind++	20.59	0.8294	0.0875	21.17	0.8814	0.0678
SGM	20.06	0.8158	0.0727	22.05	0.9089	0.4841
FECNet	20.67	0.7952	0.0995	22.57	0.8938	0.0699
HDMNet	18.55	0.7132	0.1717	20.54	0.8539	0.0690
SNR-Aware	21.48	0.8478	0.0740	24.13	0.9269	0.0318
Bread	20.83	0.8217	0.0949	17.63	0.8376	0.0681
FourLLIE	22.34	0.8403	0.0573	24.65	0.9192	0.0389
Ours	22.64	0.8589	0.0520	25.83	0.9314	0.0234

Methods	LSRW-Huawei		
	PSNR ↑	SSIM ↑	LPIPS ↓
NPE	17.08	0.3905	0.2303
LIME	17.00	0.3816	0.2069
SRIE	13.42	0.4282	0.2166
Kind	16.58	0.5690	0.2259
MIRNet	19.98	0.6085	0.2154
Kind++	15.43	0.5695	0.2366
SGM	18.85	0.5991	0.2492
FECNet	21.09	0.6119	0.2341
HDMNet	20.81	0.6071	0.2375
SNR-Aware	20.67	0.5911	0.1923
Bread	19.20	0.6179	0.2203
FourLLIE	21.11	0.6256	0.1825
Ours	21.47	0.6331	0.1781

Generalization on LSRW Dataset. To evaluate the generalization capability of our model on unseen devices, we compare the methods on the LSRW-Huawei dataset. As shown in the bottom section of Table 1, LLIE_F^2 maintains its superiority with a PSNR of 21.47 dB and SSIM of 0.6331.

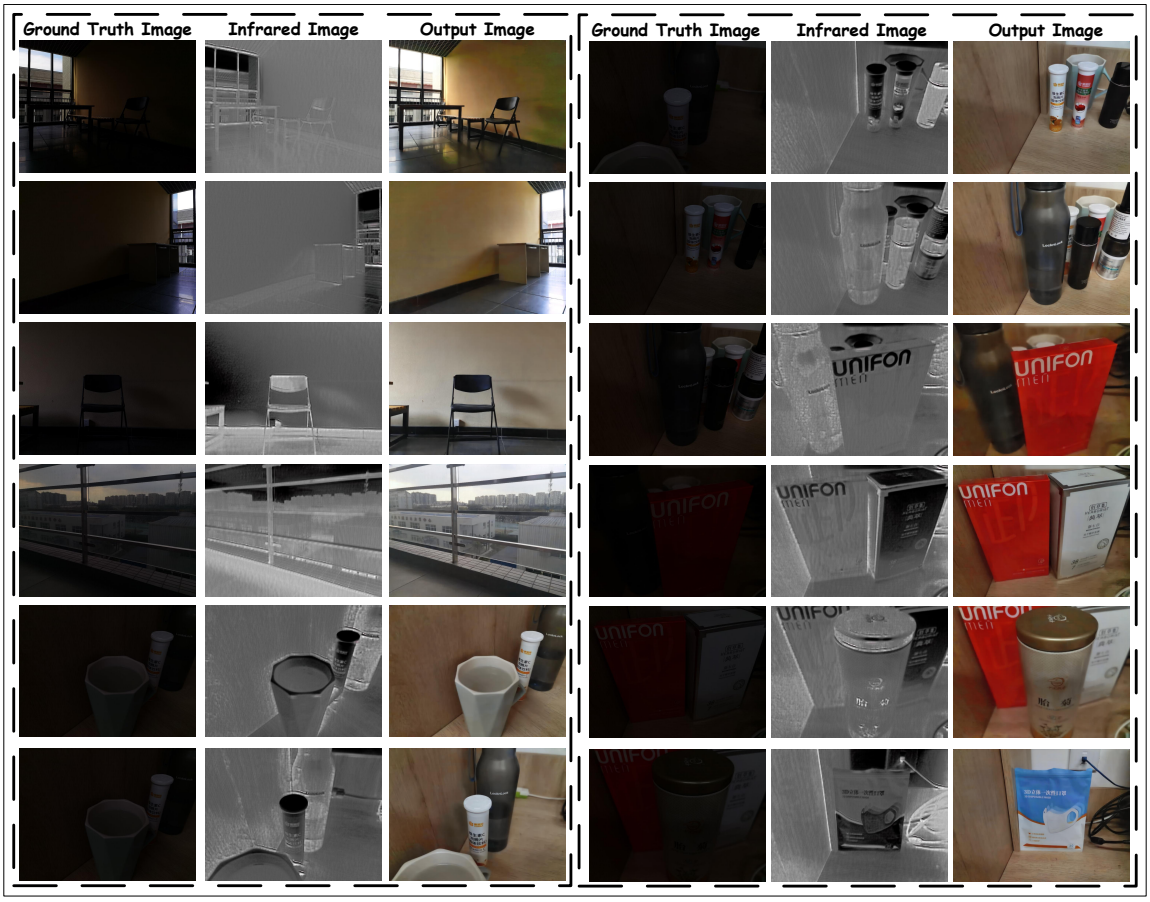


Figure 4: The performance of our **LLIE_F^2**.We present a detailed display of the input low-light images, the corresponding infrared (IR) images, and our final reconstructed outputs.

Unlike traditional methods (e.g., LIME and NPE) that suffer from severe noise amplification in unseen scenarios, our method exhibits robust adaptability. Quantitatively, our method surpasses the runner-up FourLLIE by 0.36 dB in PSNR and 0.0075 in SSIM. Even compared to the competitive FECNet, we still maintain a lead of 0.38 dB in PSNR, proving the stability of our method across different hardware sensors.

Effectiveness of Filter Design. The consistent improvement in SSIM and LPIPS metrics highlights the contribution of our specific filter designs. Existing methods often struggle to balance noise removal and detail preservation. By leveraging our designed filters in the decomposition stage, LLIE_F^2 effectively suppresses noise in the high-frequency components while preserving the structural integrity in the low-frequency components. This results in enhanced images with sharper edges and cleaner textures, as reflected by the superior quantitative scores.

Furthermore, we provide a visualization of our results to qualitatively evaluate the proposed method, As shown in Figure 4. We present a detailed display of the input low-light images, the corresponding infrared (IR) images, and our final reconstructed outputs. This section aims to demonstrate the effectiveness of our model, highlighting its remarkable performance even under extreme low-light conditions.

5.3 Albation Study

5.3.1 Quantitative experiment

To deeply investigate the contribution of each specific filter design within the LLIE_F^2 framework, we conducted a comprehensive ablation study. By systematically removing individual filtering components while keeping other parameters fixed, we analyzed their specific impacts on the final enhancement quality. As shown in Table 2, the detailed analysis of the degradation caused by the absence of each filter is presented as follows:

The removal of the Butterworth low-pass filter (W/O Filter 1) in the luminance branch introduces "patchy" artifacts. It ensures the spatial smoothness of enhancement weights; without this regularization, high-frequency noise corrupts the attention map, leading to uneven illumination and a significant drop in PSNR, which decreases from 22.64 dB to 21.05 dB on the LOL-v2-Real dataset.

The Tri-Branch Spectral Integration Filter (TBSIF, W/O Filter 2) functions as a dynamic modulation module that adaptively assigns importance weights to the features extracted from high-frequency and low-frequency filters. Specifically, it enables the model to perceive the local signal-to-noise ratio and structural complexity, thereby intelligently balancing the contribution of fine-grained textural details (high-frequency) and global structural priors (low-frequency). Without this adaptive integration mechanism, the framework loses its ability to selectively amplify salient edges while suppressing spectral noise, leading to either unintended over-smoothing or excessive noise leakage. Consequently, the SSIM on the real-world dataset drops sharply to 0.7951.

The first-order IIR filter (W/O Filter 3) in the infrared branch is vital for stabilizing cross-modal structural priors by eliminating pixel-level jitter. Without it, phase information injected into the visible branch becomes unstable, triggering structural misalignment and artifacts, particularly on sensitive datasets like LSRW-Huawei, where we observe a massive performance drop of 3.35 dB in PSNR (down to 18.12 dB).

The optimum FIR filter (W/O Filter 4) acts as a stabilizer for the complex spectrum before amplitude-phase decoupling. Omitting this pre-smoothing stage results in severe "ringing" or "checkerboard" artifacts due to frequency-domain outliers, undermining the numerical stability of the first stage and lowering the PSNR on LOL-v2-Syn by 1.81 dB.

In the texture refinement stage, the fixed high-order IIR filter (W/O Filter 5) utilizes recursive properties to stabilize local feature responses. Its removal makes it difficult to suppress noise-induced oscillations, leading to perceptually unnatural textures and a subsequent increase in LPIPS scores, which nearly doubles to 0.0955 on the LOL-v2-Real dataset.

The LMS-based adaptive FIR filter (W/O Filter 6) is critical for detail preservation as it distinguishes genuine textures from random noise. Removing it forces the model into static weight

processing, which tends to over-smooth fine details in extreme dark regions and degrade perceptual realism, evidenced by a 2.12 dB drop in PSNR on the LSRW benchmark.

The band-pass filter (W/O Filter 7) in the global branch is core to maintaining image integrity. By explicitly isolating mid-frequency structural components, it prevents the refinement process from being overwhelmed by artifacts. Experimental results confirm that its removal leads to a catastrophic drop in PSNR, yielding the lowest performance of 18.74 dB on LOL-v2-Real among all variants.

Finally, the learnable adaptive filter (W/O Filter 8) provides the flexibility to capture complex spectral patterns. Working in tandem with the band-pass filter, it enhances the model's adaptability to various sensor-specific spectral characteristics. Removing this component weakens the overall robustness, causing a consistent decline in SSIM across both synthetic and real-world datasets.

In summary, these ablation results demonstrate that the proposed filter designs are not merely redundant additions but essential components that address specific degradation problems in low-light imaging. The synergistic interaction among these diverse filters ensures that LLIE_F^2 achieves a robust balance between noise suppression, detail preservation, and structural stability. The removal of any single component disrupts this equilibrium, validating the necessity of our holistic multi-filter design strategy.

5.3.2 Visualization

In order to better demonstrate the effectiveness of our model, we also visualized the images generated by the model after removing each filter, as shown in the Figure 6 and 5. We conducted thorough ablation experiments from both quantitative analysis and qualitative visualization perspectives, and clearly discovered that the filter can perform better in enhancing low-light images.

6 Conclusion

In this report, we propose $LLIE_F^2$, an innovative dual-modality, two-stage enhancement framework designed to bridge classical digital signal processing (DSP) theory and modern adaptive architectures. By decoupling the restoration process into a "Frequency Domain Coarse Reconstruction Stage" and a "Texture Fine Reconstruction Stage," the framework effectively addresses the persistent contradiction between global illumination imbalance and local noise amplification in low-light imaging.

During the frequency domain reconstruction stage, we strategically incorporate the Butterworth Low-pass Filter and first-order IIR recursive filters to guarantee spectral stability across processing branches. On this basis, the synergy between the Tri-Branch Spectral Integration Filter (TBSIF) and the optimum FIR filter enables effective pre-smoothing of the complex spectrum and dynamic spectral balancing. Subsequently, in the texture refinement stage, high-order IIR filters and LMS-based adaptive FIR filters are further employed to enhance image clarity while preserving perceptual texture naturalness, preventing the loss of fine details. Moreover, the combination of the Band-pass Filter and learnable adaptive filters serves as the cornerstone of image integrity, bolstering the system's adaptability to diverse sensor-specific spectral characteristics. Empirical evidence confirms that the coordination of these specific filtering components is vital for maintaining structural

Table 2: Ablation study on the effectiveness of different filter configurations. We evaluate the contribution of each component by systematically removing the corresponding filters labeled in Figure 2 and Figure 3. The variants W/O Filter 1–8 refer to the models without: (1) Butterworth Low-pass Filter, (2) TBSIF, (3) First-order IIR Filter, (4) Optimum FIR Filter, (5) Fixed high-order IIR, (6) FIR filtering via LMS, (7) Band-Pass Filter, and (8) Learnable Adaptive Filter. "Ours" represents the full model incorporating all designs. Best results are highlighted in red.

Methods	**LOL-v2-Real**			**LOL-v2-Syn**		
	PSNR ↑	**SSIM** ↑	**LPIPS** ↓	**PSNR** ↑	**SSIM** ↑	**LPIPS** ↓
W/O Filter 1	21.05	0.8142	0.0814	23.41	0.8922	0.0514
W/O Filter 2	19.88	0.7951	0.1123	22.15	0.8756	0.0782
W/O Filter 3	22.12	0.8420	0.0615	24.89	0.9188	0.0341
W/O Filter 4	21.45	0.8299	0.0742	24.02	0.9015	0.0456
W/O Filter 5	20.67	0.8012	0.0955	22.88	0.8844	0.0621
W/O Filter 6	21.98	0.8456	0.0598	25.12	0.9231	0.0305
W/O Filter 7	18.74	0.7633	0.1450	20.95	0.8522	0.1023
W/O Filter 8	22.31	0.8510	0.0552	25.40	0.9267	0.0274
Ours	22.64	0.8589	0.0520	25.83	0.9314	0.0234

Methods	**LSRW-Huawei**		
	PSNR ↑	**SSIM** ↑	**LPIPS** ↓
W/O Filter 1	19.82	0.5814	0.2241
W/O Filter 2	20.45	0.5956	0.2112
W/O Filter 3	18.12	0.5120	0.2785
W/O Filter 4	21.01	0.6215	0.1894
W/O Filter 5	20.14	0.5899	0.2205
W/O Filter 6	19.35	0.5562	0.2456
W/O Filter 7	21.22	0.6288	0.1821
W/O Filter 8	20.78	0.6041	0.2033
Ours	21.47	0.6331	0.1781

integrity.

Extensive evaluations on benchmark datasets, including LOL-v2-Real, LOL-v2-Syn, and LSRW-Huawei, demonstrate that $LLIE_F^2$ achieves state-of-the-art (SOTA) performance compared to recent methodologies. This work highlights the significant potential of embedding classical DSP filtering theories within adaptive systems to enhance both stability and interpretability.

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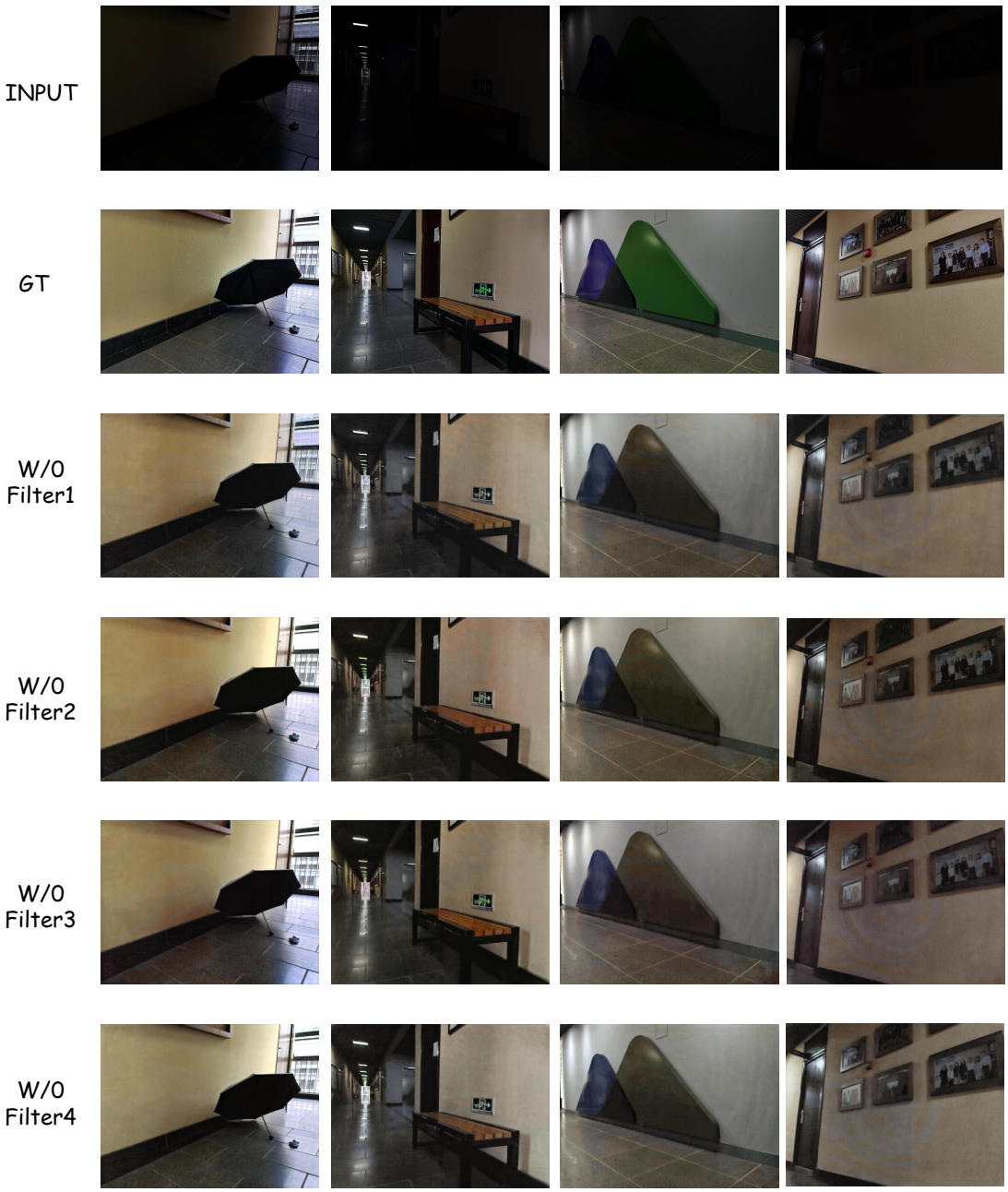
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INPUT
GT
W/O Filter1
W/O Filter2
W/O Filter3
W/O Filter4

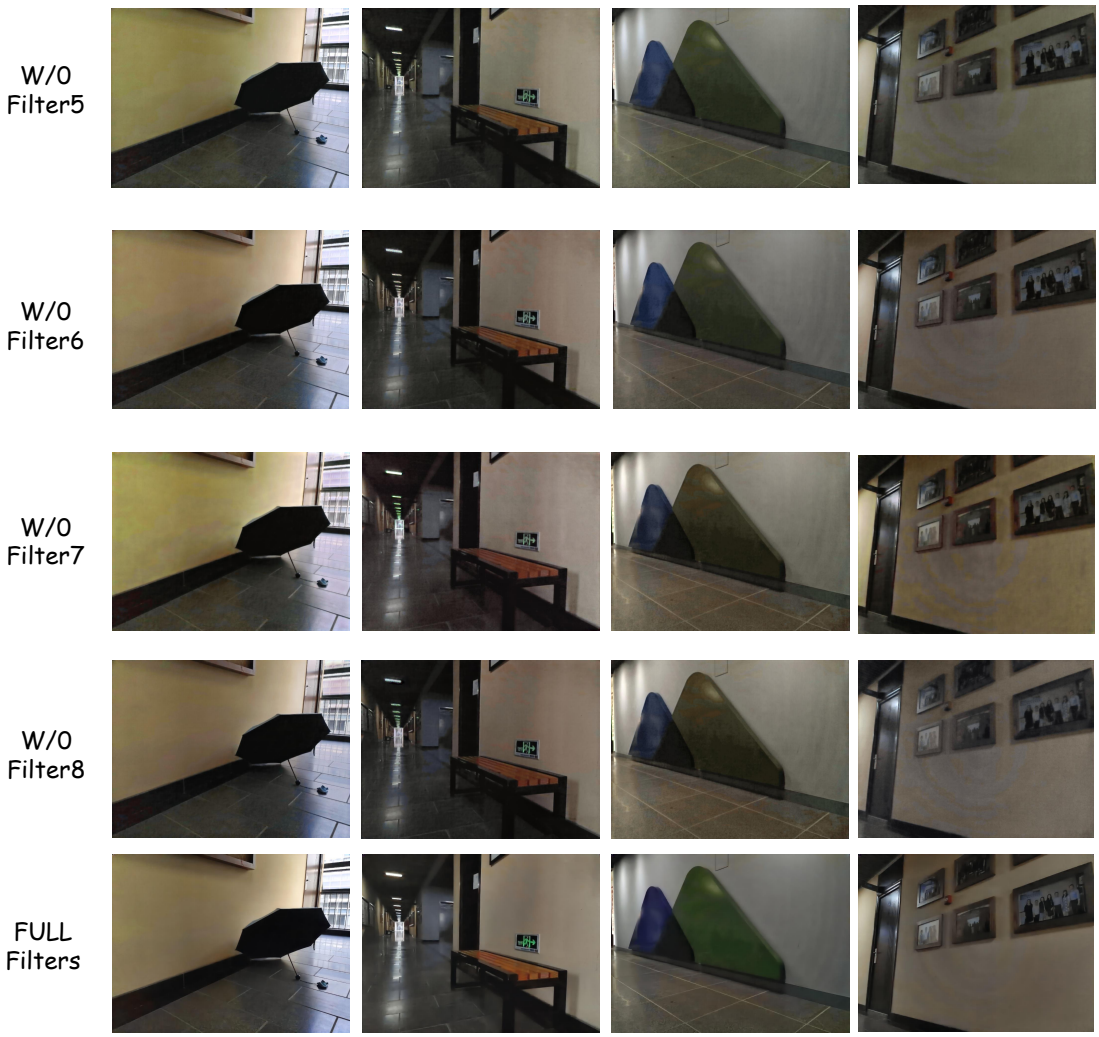


Figure 6: The Albation Study for Filters in **LLIE_F**^{2} (continue).